CS 412 Intermediate Project Report
Team Members: Hannan Shahid, Yashashwi Singh

1. **Description of the data (Yash)**

For our project we will be using the Cassava Leaf Disease Classification dataset from Kaggle as this dataset has/contains 21,397 images of cassava leaves that were collected from the fields in Uganda and each of the image is labeled as either a healthy leaf or one of the four different cassava diseases. Our goal is to use these images to train and compare the different machine learning models that can identify the disease from a picture of a cassava leaf. One of the main challenges with this dataset is that the images are not equally distributed among all the five classes and for example, Cassava Mosaic Disease (CMD) makes up approximately 61 percent of the dataset and while the rarest disease class only makes up about 5 percent of the dataset. This creates a problem because a machine learning model will become better at predicting the majority class while having difficulty identifying the diseases that have fewer/lesser images. This is important for our project because we are not only trying to achieve/get high overall accuracy but we also want our models to be able to identify the rare diseases correctly as to address this problem we plan to compare the different approaches for handling the imbalanced in the dataset, focal loss, including oversampling, and class balanced loss. We will use the images as an input for our machine learning models and then compare their performance using accuracy, recall for each disease, confusion matrices, and macro f1 score as this will help us understand how well each model performs across all the five classes rather than only focusing on the overall accuracy.

2: Introduction and Problem Description (Hannan)

Cassava is an important food crop in Africa, and diseases can reduce its harvest. Identifying a disease from a leaf often requires expert knowledge. A model that recognizes disease symptoms from a photograph could help make this process more accessible.

In this project, we treat disease recognition as an image classification problem. Each input is a cassava leaf image, and the model predicts one of five classes: Cassava Bacterial Blight (CBB), Cassava Brown Streak Disease (CBSD), Cassava Green Mottle (CGM), Cassava Mosaic Disease (CMD), or Healthy.

The main challenge we study is the uneven distribution of images across these classes. CMD accounts for approximately 61.5% of the dataset, while CBB accounts for only 5.1%. A model could therefore achieve 61.5% accuracy by predicting CMD for every image, while missing every example from the other four classes. This shows why overall accuracy alone is not enough to evaluate our models [1, 2].

Figure 1: CMD accounts for 61.5% of images and has about 12 times as many images as CBB.

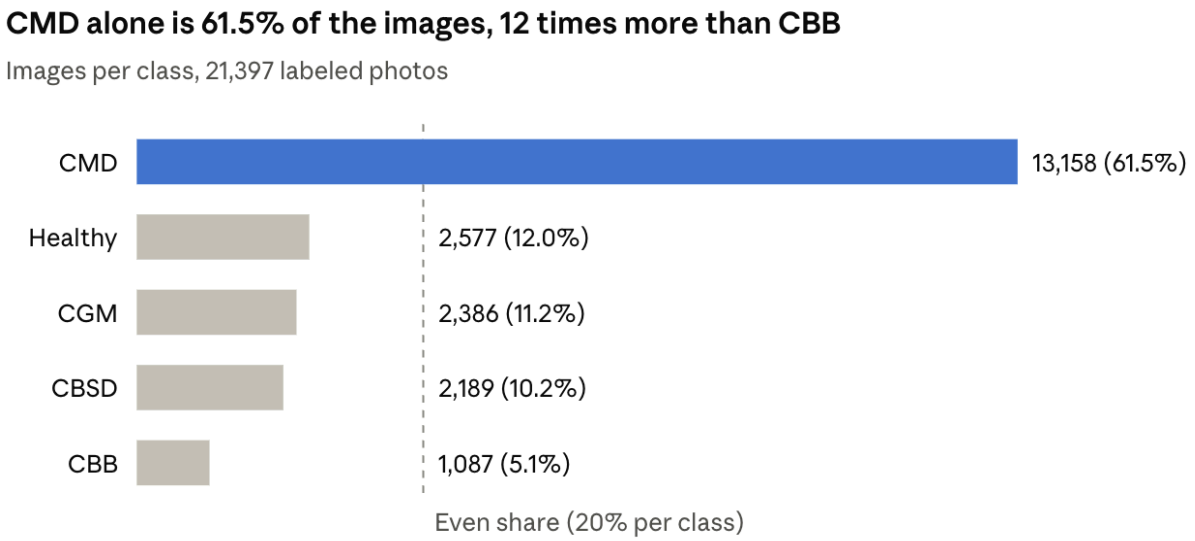


Figure 1. Class distribution of the 21,397 labeled cassava leaf images. CMD contains approximately 12.1 times as many images as CBB. The dashed line marks an equal 20% share across the five classes [2].

Earlier studies have used deep learning to recognize plant diseases, including cassava diseases [3, 4]. Our project focuses on comparing models and examining whether methods for handling class imbalance improve recognition of the less common classes.

We first compare a classical model using HOG and color features with an SVM against two CNNs, ResNet-50 and EfficientNet-B0 [7–9]. We then select the CNN with the highest validation macro F1 and compare four training approaches: ordinary training, oversampling, class-balanced loss, and focal loss.

Our main paper, by Cui et al. [5], introduces class-balanced loss based on the effective number of samples. The main idea is that additional similar images provide less new information as a class grows. The method accounts for this when calculating class weights. Its parameter β controls the strength of the weighting. Focal loss uses a different approach by reducing the contribution of easy examples so that training places more attention on difficult examples [6].

Our main evaluation measures are macro F1, which gives each class equal importance, and recall for each disease. We also report overall accuracy and confusion matrices to understand which classes the models confuse.

3: What We Have Done So Far (Hannan)

So far, our work has focused on reviewing the dataset, designing fair model comparisons, and preparing the experiments for handling class imbalance.

We examined the number of images in each class and confirmed the large difference between CMD and the smaller classes. As Figure 1 shows, CMD is the only class above an equal 20% share. This supports our decision to examine performance separately for each class.

We selected a stratified 80/10/10 split with a fixed random seed. This means approximately 80% of the images will be used for training, 10% for validation, and 10% for testing, while keeping similar class proportions in each set. Every model will use the same split. The test set will contain only about 109 CBB images, so small changes in CBB recall will need careful interpretation. We will use the validation set to select model settings and reserve the test set for the final comparison.

We also calculated a simple reference baseline. A classifier that always predicts CMD would achieve approximately 61.5% accuracy, zero recall for the other four classes, and a macro F1 of approximately 0.152. These values were calculated from the dataset's class counts; they are not results from a trained model.

For the classical baseline, we designed a pipeline that resizes images to 128 × 128 pixels and combines HOG features with color histograms before passing them to an SVM. HOG captures edges and local shapes, while color histograms describe the distribution of colors in an image. We will select the SVM settings using the validation set.

For ResNet-50 and EfficientNet-B0, we selected ImageNet-pretrained weights and a final classification layer with five outputs. We plan to use comparable training budgets and the same augmentation approach, including random crops, flips, and small color changes. Augmentation will be applied only to training images. Training and evaluating these models remain the next stage of our work.

I also reviewed the class-balanced loss method and calculated example class weights using the full dataset counts. After normalizing the five weights to sum to five, $\beta = 0.9$ and $\beta = 0.99$ produce almost equal weights. At $\beta = 0.999$, CBB receives approximately 1.51 times the weight of CMD. At $\beta = 0.9999$, this ratio increases to approximately 7.10.

Figure 2: the line graph showing class-balanced weights at different β

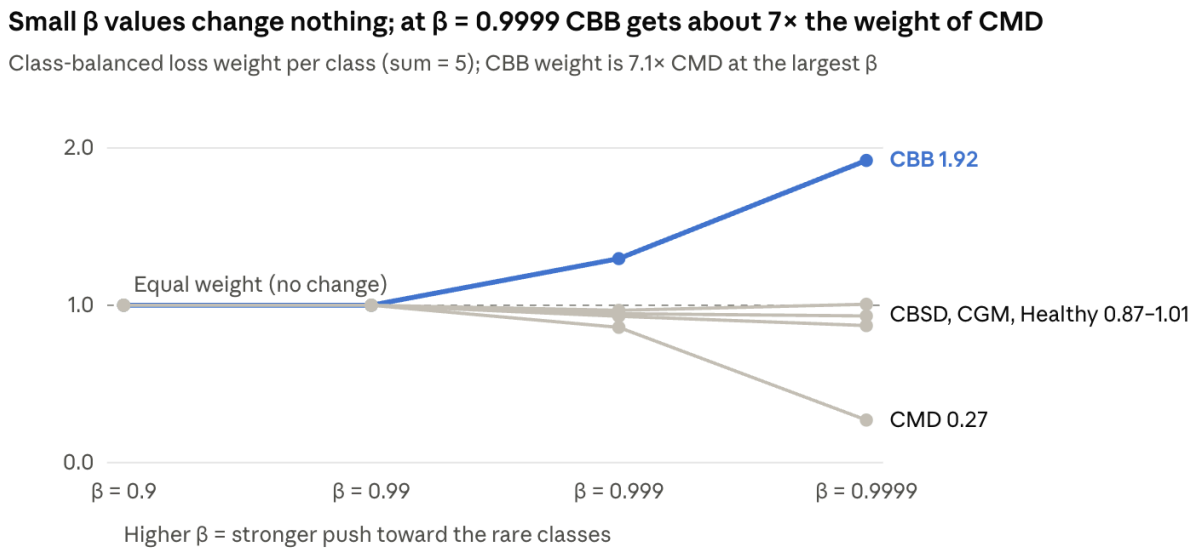


Figure 2. Example class-balanced loss weights calculated using the full dataset counts and normalized to sum to five. At β = 0.9999, the CBB weight is approximately 1.92 and the CMD weight is approximately 0.27, giving a ratio of about 7.1.

These calculations help us choose settings for the experiments, but they do not establish whether stronger weighting improves predictions. I plan to compare β = 0.999 and β = 0.9999 using validation macro F1 and recall for each disease. For actual training, I will recalculate the weights using only the training-set counts. Oversampling will also be applied only to the training set.

My work has focused on reviewing the imbalance methods and preparing the analysis for each disease. Yashashwi has focused on data preparation and the baseline model designs. We will work together on comparing the results, preparing the presentation, and completing the final report.

4. What remains to be done (Yash)

For the remaining parts of our project we will be focusing on completing our model comparisons and testing the different methods to improve the identification of the rare cassava leaf diseases. We will first be making sure that our classical baseline model is using HOG and SVM as well as/along with our two CNN models, EfficientNet-B0, and ResNet-50 have been trained and have been evaluated so that we can compare their performances. After comparing these models we will focus on the handling of the class imbalance in the dataset as we plan to test the three different approaches that are balanced loss, oversampling, and focal loss. Oversampling will allow the model to see images from the rare disease classes more frequently during the

training. Class balanced loss will give more importance to classes with the fewer images and lastly, while focal loss will help the model focus more on the examples that are difficult to classify. Once we have completed all these experiments we will compare the results using confusion matrices, macro F1 score, accuracy, and recall for each disease as this will help us see which approach performs better and especially/particularly for the rare diseases. According to our project schedule we plan to complete these experiments between 2nd November and November 14th. After that we will be working together on preparing our final presentation, that will be including a possible demonstration use a cassava leaf image and lastly, finally we will organize our findings, discuss the results among ourselves, and complete our final project report by December 1st 2026.

References

[1] Makerere University AI Lab and Kaggle. 2020. Cassava Leaf Disease Classification.

[2] Y. Zhong, B. Huang, and C. Tang. 2022. Classification of Cassava Leaf Disease Based on a Non-Balanced Dataset Using Transformer-Embedded ResNet. *Agriculture*, 12(9), 1360.

[3] S. P. Mohanty, D. P. Hughes, and M. Salathé. 2016. Using Deep Learning for Image-Based Plant Disease Detection. *Frontiers in Plant Science*, 7, 1419.

[4] A. Ramcharan et al. 2017. Deep Learning for Image-Based Cassava Disease Detection. *Frontiers in Plant Science*, 8, 1852.

[5] Y. Cui, M. Jia, T.-Y. Lin, Y. Song, and S. Belongie. 2019. Class-Balanced Loss Based on Effective Number of Samples. *CVPR*, 9268–9277.

[6] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár. 2017. Focal Loss for Dense Object Detection. *ICCV*, 2980–2988.

[7] N. Dalal and B. Triggs. 2005. Histograms of Oriented Gradients for Human Detection. *CVPR*, 886–893.

[8] K. He, X. Zhang, S. Ren, and J. Sun. 2016. Deep Residual Learning for Image Recognition. *CVPR*, 770–778.

[9] M. Tan and Q. V. Le. 2019. EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. *ICML*, 6105–6114.